THE FINITE HILBERT STACK OF AN ALGEBRAIC SPACE IS ALGEBRAIC

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Representability on top. lemma-representable-on-top (same chapter, named). H_d(X) -> H_d is representable by algebraic spaces.
- Algebraicity of H_d. proposition-finite-hilbert-point (same chapter, named). H_d is algebraic.
- Representable to algebraic. algebraic-lemma-representable-morphism-to-algebraic (cross-chapter, named). Concludes.

Source reference. Original reference 05UM: focused-seeds/geometry/S9-crit-hilbert-stack-of-space/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05YJ: sources/stacks-criteria/pinned-original-source.tex, source lines 2178–2178. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original proposition 05YQ: sources/stacks-criteria/pinned-original-source.tex, source lines 2485–2485. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Then $\mathcal{H}_d(X)$ is an algebraic stack.*

Proof. The 1-morphism

$$\mathcal{H}_d(X) \longrightarrow \mathcal{H}_d$$

is representable by algebraic spaces according to Lemma 05YJ. The stack $\mathcal{H}_d$ is an algebraic stack according to Proposition 05YQ. Hence $\mathcal{H}_d(X)$ is an algebraic stack by Algebraic Stacks, Lemma 05UM. □

Date: stacks-project:criteria.tex:local-snapshot-20261012; GNU FDL 1.2 or later.